

Appendix D: Prompt Templates, Output Schema and RAG Configuration

D.1 Purpose

Appendix D records the reusable prompt templates, the shared structured-output schema and the locked RAG configuration. It supports fair comparison by holding the task, label schema, evidence boundary and output structure consistent across both context-delivery conditions.

Note: In this appendix, `condition` is retained as the implementation and prompt-template field name used in the locked artefacts. It denotes the executed context-delivery strategy: `Long_Context` for full-filing context delivery and `RAG` for the RAG-based strategy.

D.2 Shared Task Boundary

The shared task boundary requires each model response to classify the claim using only the filing evidence supplied in that condition. In the `Long_Context` condition, the supplied evidence is the full cleaned SEC 10-K filing text. In the `RAG` condition, the supplied evidence is the locked set of retrieved chunks from the same cleaned filing corpus. The model-facing instruction excludes external knowledge, market knowledge, memory, investment judgement and assumptions outside the supplied filing evidence. The model is instructed to return *Insufficient Evidence* when the supplied evidence is missing, incomplete, ambiguous, too general or insufficiently specific.

D.3 Long-Context Prompt Template: LC_PROMPT_v1.0-final

[SYSTEM INSTRUCTION]

You are evaluating SEC 10-K filing evidence for a controlled academic experiment. Classify the claim using only the provided filing evidence. Do not use outside knowledge. Return only valid JSON matching the required schema.

[USER TEMPLATE]

Experiment condition: `Long_Context`

Case ID: `{case_id}`

Filing ID: `{filing_id}`

Company: `{company}`

Report date: `{report_date}`

Claim to classify:

`{claim_text}`

Allowed labels:

- Supported: the filing evidence supports the claim.
- Contradicted: the filing evidence conflicts with the claim.
- Insufficient Evidence: the provided filing evidence does not establish the claim.

Full cleaned SEC 10-K filing evidence:

`{context_text}`

Return JSON only.

D.4 RAG-Based Prompt Template: RAG_PROMPT_v1.0-final

[SYSTEM INSTRUCTION]

You are evaluating SEC 10-K filing evidence for a controlled academic experiment. Classify the claim using only the provided filing evidence. Do not use outside knowledge. Return only valid JSON matching the required schema.

[USER TEMPLATE]

Experiment condition: RAG

Case ID: {case_id}

Filing ID: {filing_id}

Company: {company}

Report date: {report_date}

Claim to classify:

{claim_text}

Allowed labels:

- Supported: the filing evidence supports the claim.
- Contradicted: the filing evidence conflicts with the claim.
- Insufficient Evidence: the provided filing evidence does not establish the claim.

Retrieved SEC 10-K filing evidence chunks:

{context_text}

Return JSON only.

D.5 JSON Output Schema: sec10k_claim_classification_schema_v1.0-final

Table D.5 records the shared structured-output fields required from both experimental conditions.

Table D.5: JSON Output Schema for Claim Classification Output

Field	Type	Requirement
case_id	String	Must match the input claim-case identifier.
predicted_label	Enum	Must be one of Supported, Contradicted, or Insufficient Evidence.
evidence_references	Array	Array of evidence-reference objects used for the decision. Each item contains reference and evidence_role.
evidence_references[].reference	String	Filing, section, chunk, excerpt or other supplied-context reference used by the model.
evidence_references[].evidence_role	String	Short description of how the referenced evidence supports the classification decision.
reasoning_summary	String	Concise explanation tied to the supplied SEC 10-K filing context.
confidence	Enum	High, Medium, or Low; used descriptively, not as a scoring substitute.

D.6 RAG Configuration: RAG_CONFIG_v1.0-final

Table D.6 records the locked retrieval configuration used to construct the RAG evidence supplied to the model.

Table D.6: Locked RAG Configuration

Configuration item	Locked rule	Rationale
Source corpus	Same cleaned SEC 10-K filing text used in the Long_Context condition.	Keeps the comparison focused on context-delivery strategy rather than source differences.
Chunk size	750 words.	Provides enough surrounding context for financial-disclosure passages while keeping retrieved evidence smaller than the full filing.
Chunk overlap	100 words.	Reduces boundary loss where relevant evidence spans the end of one chunk and the start of the next.
Top-k	8 chunks per claim-case.	Gives the RAG condition multiple evidence opportunities without turning it into a near-full-document condition.
Retrieval query	Natural-language claim text from the locked model-call queue only; expected label, hidden rationale and target evidence snippets are excluded.	Prevents answer-key leakage and keeps retrieval based on the same claim available to both conditions.
Metadata retained	RAG context records retain run_id, case_id, filing_id, top_k, embedding_model and retrieved_chunks. Retrieved chunks retain rank, chunk identifier, word range, retrieved text and similarity score.	Supports reproducibility and post-run retrieval-error diagnosis.
Manual retrieval changes after lock	Not allowed. Retrieval-completeness failures are not corrected post hoc and are considered during qualitative error analysis where applicable.	Prevents outcome-driven retrieval repair after model output is known.
Retrieval implementation	Preparation-time in-memory cosine-similarity ranking per filing, with persisted chunk and locked context artefacts.	Matches the implemented preparation path without implying FAISS or managed-vector-store use; scored execution loaded the pre-generated context by run_id.
Embedding model	text-embedding-3-large.	Provides a fixed retrieval baseline so differences are not caused by changing embedding models.

Configuration item	Locked rule	Rationale
Context artefacts persistence	Saved RAG context artefacts retain the retrieved chunk identifiers, word ranges, scores and text supplied to the RAG condition.	Allows the retrieved evidence actually supplied to each RAG call to be audited later.
Scored-execution context path	Load the pre-generated and locked context for the corresponding RAG run_id; do not rerun retrieval during scored model execution.	Separates retrieval preparation from the controlled model-call path and prevents post-lock retrieval drift.